

AI:

Artificial Intelligence

→ First topic for paper preparation:-

Knowledge Representation:-

→ machine ko intelligent banany k lie, to human knowledge ikthi kofa h diff resources se or intelligence kia h ability to use that knowledge. Representation kofa phr reasoning kofa (at right time) →
ye sab machine ko seekhana pory ga 😊

Correct input → Correct Output

if we don't represent correctly → to syntax error ayn gyn, semantic error (meaning galt).

Methods of giving knowledge :-

Sab se phly hm uski logic
dkhy gyh.
logic :

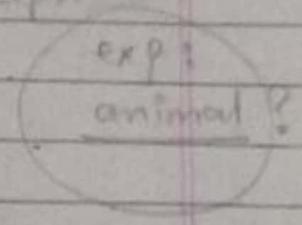
① ~~Syntax~~
prepositional
logic
true / false

② ~~Semantic~~
predicate
logic
quantifiers

Rules: →
If / then

→ E / V
Deductive reasoning, data view con.
Forward chaining
Backward chaining
Conflict resolution

Semantic net. → meaningful
→ graph
google graph



Frames

Slots / Fillers

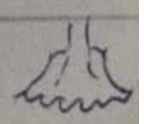
↓ ↓
obj attribute

Script:

advance form of frames
like movies.

Natural language processing:- (NLP)

- How human communicate through diff. languages with each other.
- Computer should replicate the same thing.



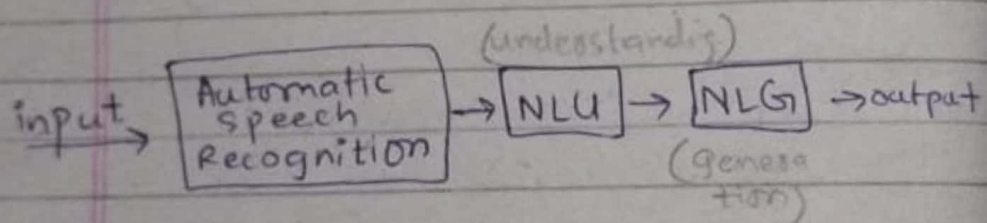
Applications of NLP:

- ① Speech Recognition
 - ① google assistant
 - ① Siri
 - ① Alexa

- ② Sentimental Analysis
 - ① Tweets analysis
 - ① Content analysis

- ③ Machine Translations:
 - ① language translation

- ④ Chatbots:
 - ① robots



NLU:

What do the users say?
their intent? meaning?

Challenges: $\downarrow$ all word k multiple meaning.

- $\rightarrow$ lexical ambiguity
- $\rightarrow$ Syntactic Ambiguity
- $\rightarrow$ Semantic " $\rightarrow$ means klat?
- $\rightarrow$ Pragmatic $\rightarrow$ context k multiple meaning for machine $\rightarrow$

NLG:

- after understanding
- What should we say to user?
- It should be intelligent and conversational.
- Deal it structured data
- Text / Sentence planing.

Propositional logic :-

→ (True / False, not both)

PL
^
Syntax Semantic (meanings)

Atomic single Proposition Symbol used	Complex / Composite → do sentence mill kx aik proposition bnate hn. Symbols :- - Negation (not) V Disjunction (or) ^ Conjunction (and) → if then ↔ iff (done side)
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Predicate logic:

20 is smaller than 10
 if $\rightarrow$ x is even no.
 (involving ① constants
 ② variables)

③ Quantifiers: ④ logical connectivity

Universal Quantified

($\forall$)

($\forall$)

x is < 10

P(x)

$\Phi(x, y)$

P(x)

P(1) $\rightarrow$ T

P(15) $\rightarrow$ F

Existential Quantified

($\exists$)

($\exists$)

$\Phi(x)$

Domain: {1, 3, 5}

$\exists x \Phi(x)$

→ aik bhi true to true hujy ga

$\forall x P(x) = \text{True}$

Domain Value of x

Domain: {1, 3, 5, 7}

False value

False

True value

True

Examples

Not All Glitters are gold.

Semantic Network:

"Graphical notation for representing knowledge in interconnected nodes pattern"

- Popular in AI and NLP
- Alternative of predicate logic
- Nodes represents objects
 - circle (Physical object)
 - ellipse (concept)
 - rectangle (situation)

- Arcs represent relation with objects (links, arrows)
- link labels specify relationships
- Also known as associative nets (nodes are associated).

Components :-

i) lexical Components

Nodes, Links, Labels

ii) structural

link or Nodes → directed

(iii) Semantic

Definition related to link & nodes → basically facts

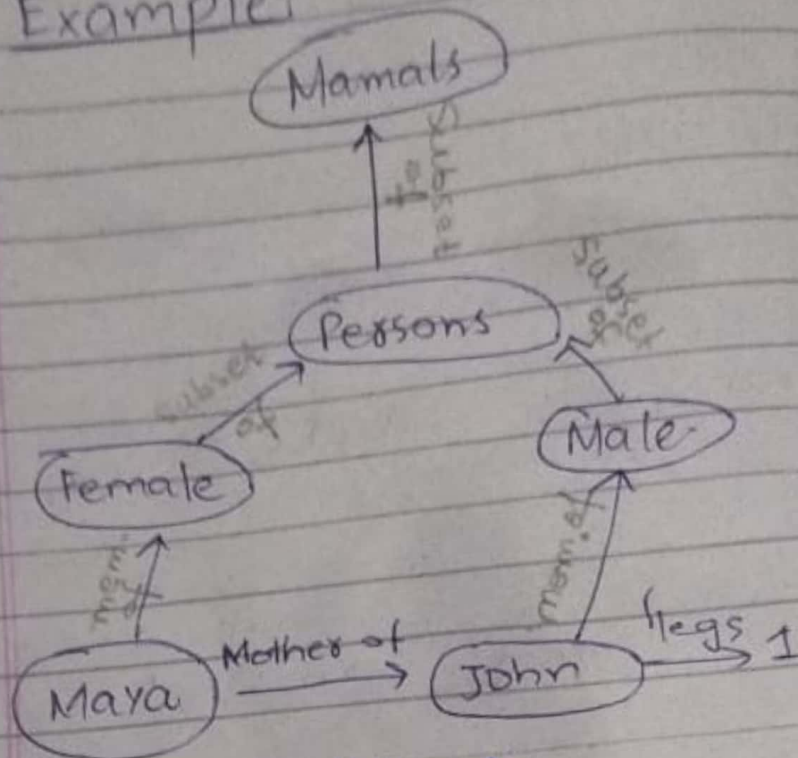
(iv) Procedural

i) constructor → Creation of links

ii) Distructor → Removal of link

- ① Nodes ② Links
③ Link labels.

Example:



Scripts: / objects:-

→ it is a structure that prescribes a set of circumstances which could be follow from one another.

→ Considered of consist of a no. of slots or frames. but with more specialized roles

script

Components:

Roles:

Person involved in event

ex: student

Structure:

object

- ① Object Name
- ② Properties
- ③ Methods

Props:

→ Objects involve in event exp: pen

Entry Conditions:

→ Condition that needs to be satisfied before event occurs in script.

Results:

→ Condition that will be true after event in script. exp: roll no. slip

Track:

→ Variation are the script. exp: fill ppr

Scenes:

→ Sequence of event that occurs. exp: diff share ppr

Adv

① Event Prediction is possible

disadv

→ less general than frames

② Single coherent interpretation may be build up of collection of observation

→ May not be suitable to represent all kind of knowledge

- ① Frame Name
- ② Slots
- ③ Facets
- (in oo way)

Frames:

"Frames is a collection of attributes or slots and associated values that describe some real-world entity."

- Uses data structure (records) to represent the knowledge represented in semantic network
- each frame represents the node in semantic network as a class, or an instance and each relation as slot.

Attributes of Slots :-

- ① Instance
- ② Definition
- ③ Default
- ④ Domain
- ⑤ Range
- ⑥ Range constraints
- ⑦ TO - compute
- ⑧ Single-valued
- ⑨ Inverse
- ⑩ Transfer - through

Reasoning action that
can be performed Frames:

- (1) Relating the definition.
- (2) Inheritance
- (3) Locality of value:
(valid or not)
- (4) Consistency check
- (5) Maintaining consistency.
- (6) Computation of value of slot.

example

Frame : Car

Slots : model, color

Facets : model : 2001, 2002
color : Green, Purple etc